Discrete Mathematics

CSE 4203

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6.2 Pigeonhole Principle

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Introduction

- The **Pigeonhole Principle (PHP)** is a fundamental combinatorial principle.
- Informally: If more objects (pigeons) are placed into fewer containers (pigeonholes), at least one container has more than one object.
- Useful in proofs involving counting, existence, and combinatorics.

Basic Pigeonhole Principle

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- Example: If 367 people are in a room, at least two share the same birthday (ignoring leap years).

Generalized Pigeonhole Principle

Theorem (Generalized PHP)

If N objects are placed into k boxes, then at least one box contains at least

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- Example: If 100 balls are placed in 9 bins, one bin has at least $\lceil 100/9 \rceil = 12$ balls.

Applications of PHP

- Existence proofs in number theory.
- Proving bounds in combinatorics and graph theory.
- Guaranteeing repetition or duplication in large data sets.
- Useful in algorithm analysis.

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Solution:

- There are 12 months (pigeonholes) and 13 people (pigeons).
- By PHP, at least one month has at least $\lceil 13/12 \rceil = 2$ people.

Harder Example

Problem: Show that in any set of 100 integers, there exist two whose difference is divisible by 99.

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Solution:

- Consider the remainders when dividing each integer by 99. There are 99 possible remainders.
- With 100 integers (pigeons) and 99 remainders (holes), at least two numbers share the same remainder.
- Their difference is divisible by 99.

Example: Pigeonhole Principle in Grading

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$$\text{5 students for each grade (A, B, C, D, F).}$$

- With 26 students, at least one grade must be assigned to at least 6 students.

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- With 26 students, at least one grade must be assigned to at least 6 students.

Answer:

Minimum number of students = 26.

Example: Pigeonhole Principle with Playing Cards

Problem:

a. How many cards must be selected from a standard deck of 52 cards to guarantee that at least **three cards of the same suit** are selected?
b. How many cards must be selected to guarantee that at least **three hearts** are selected?

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Solution:

(a) There are 4 suits (pigeonholes). To avoid having 3 of the same suit, you could have at most 2 cards from each suit: $2 \times 4 = 8$ cards.

$$\text{Select one more card: } 8 + 1 = 9.$$

So, **9 cards** guarantee 3 of the same suit.

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b. For hearts, there are 13 hearts in the deck. Worst case: select all 39 non-hearts first.

$$39 + 3 = 42.$$

So, **42 cards** guarantee 3 hearts.

Example: Consecutive Days with Fixed Total Games (1/2)

Problem: During a month with 30 days, a baseball team plays at least one game each day, but no more than 45 games in total. Show that there must be a period of some number of consecutive days during which the team plays **exactly 14 games**.

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Setup:

- Let g_i = number of games played on day i, $1 \leq i \leq 30$.
- Define prefix sums:

$$s_0 = 0, \quad s_k = g_1 + g_2 + \cdots + g_k \quad (1 \leq k \leq 30).$$

- There are 31 prefix sums: $s_0, s_1, \ldots, s_{30}$.

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- Construct two sets of integers:

$$t_i = s_i, \quad u_i = s_i + 14.$$

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- This means there exists a consecutive block of days where the team played **exactly 14 games**.

Pigeonhole Principle Example

Problem: How many numbers must be selected from the set $\{1, 2, 3, 4, 5, 6\}$ to guarantee that at least one pair of these numbers adds up to 7?

Solution:

- Partition the set into complementary pairs that sum to 7:

$$\{1, 6\}, \quad \{2, 5\}, \quad \{3, 4\}.$$

  These are 3 "pigeonholes."
- To avoid having any pair that sums to 7, one can pick at most one element from each pair. Thus the maximum number of elements you can choose without creating such a pair is 3 (pick one from each pair, e.g. $\{1, 2, 3\}$).
- Therefore, selecting one more element forces two chosen numbers to lie in the same pair, producing a pair that sums to 7.

Answer

You must select at least 4 numbers to guarantee that some chosen pair sums to 7.

Summary

- PHP is a powerful counting argument.
- Basic: More objects than boxes $\Rightarrow$ some box has ≥ 2 objects.
- Generalized: At least one box has $\lceil N/k \rceil$ objects.
- Frequently used in number theory, combinatorics, and computer science.